Topic: Divergence Theorem

I. Introduction

To motivate this section, lets start with a problem

Compute the flux of $\mathbf{F} = [x^2, 0, z^2]$ through the rectangular box: $|x| \leq 1$, $|y| \leq 3$, and $0 \leq z \leq 2$. Let S be the surface of the box.

Note: Since S is composed of 6 surfaces, we need to compute the flux through each surface.

$$\begin{aligned} Flux &= \iint\limits_S \mathbf{F} \cdot \mathbf{n}\, dA \\ &= \iint\limits_{S_{x-}} \mathbf{F} \cdot \mathbf{n}_{x-}\, dA + \iint\limits_{S_{x+}} \mathbf{F} \cdot \mathbf{n}_{x+}\, dA + \iint\limits_{S_{y-}} \mathbf{F} \cdot \mathbf{n}_{y-}\, dA + \iint\limits_{S_{y+}} \mathbf{F} \cdot \mathbf{n}_{y+}\, dA + \iint\limits_{S_{z-}} \mathbf{F} \cdot \mathbf{n}_{z-}\, dA + \iint\limits_{S_{z+}} \mathbf{F} \cdot \mathbf{n}_{z+}\, dA \end{aligned}$$

S_{x-}: $x = -1$, $-3 \leq y \leq 3$, $0 \leq z \leq 2$, and $\mathbf{n}_{x-} = [-1, 0, 0]$

S_{x+}: $x = 1$, $-3 \leq y \leq 3$, $0 \leq z \leq 2$, and $\mathbf{n}_{x+} = [1, 0, 0]$

S_{y-}: $y = -3$, $-1 \leq x \leq 1$, $0 \leq z \leq 2$, and $\mathbf{n}_{y-} = [0, -1, 0]$

S_{y+}: $y = 3$, $-1 \leq x \leq 1$, $0 \leq z \leq 2$, and $\mathbf{n}_{y+} = [0, 1, 0]$

S_{z-}: $z = 0$, $-1 \leq x \leq 1$, $-3 \leq y \leq 3$, and $\mathbf{n}_{z-} = [0, 0, -1]$

S_{z+}: $z = 2$, $-1 \leq x \leq 1$, $-3 \leq y \leq 3$, and $\mathbf{n}_{z+} = [0, 0, 1]$

$$Flux = \iint\limits_{S_{x-}} \mathbf{F}\cdot\mathbf{n}_{x-}\,dA + \iint\limits_{S_{x+}} \mathbf{F}\cdot\mathbf{n}_{x+}\,dA + \iint\limits_{S_{y-}} \mathbf{F}\cdot\mathbf{n}_{y-}\,dA + \iint\limits_{S_{y+}} \mathbf{F}\cdot\mathbf{n}_{y+}\,dA + \iint\limits_{S_{z-}} \mathbf{F}\cdot\mathbf{n}_{z-}\,dA + \iint\limits_{S_{z+}} \mathbf{F}\cdot\mathbf{n}_{z+}\,dA$$

$$\iint\limits_{S_{x-}} \mathbf{F}\cdot\mathbf{n}_{x-}\,dA = \iint\limits_{S_{x-}} [x^2,0,z^2]\cdot[-1,0,0]\,dA = \iint\limits_{S_{x-}} -x^2\,dA \overset{x=-1}{=} \iint\limits_{S_{x-}} -1\,dA = -Area(S_{x-}) = -12$$

$$\iint\limits_{S_{x+}} \mathbf{F}\cdot\mathbf{n}_{x+}\,dA = \iint\limits_{S_{x+}} [x^2,0,z^2]\cdot[1,0,0]\,dA = \iint\limits_{S_{x+}} x^2\,dA \overset{x=1}{=} \iint\limits_{S_{x+}} 1\,dA = Area(S_{x+}) = 12$$

$$\iint\limits_{S_{y-}} \mathbf{F}\cdot\mathbf{n}_{y-}\,dA = \iint\limits_{S_{y-}} [x^2,0,z^2]\cdot[0,-1,0]\,dA = \iint\limits_{S_{y-}} 0\,dA = 0$$

$$\iint\limits_{S_{y+}} \mathbf{F}\cdot\mathbf{n}_{y+}\,dA = \iint\limits_{S_{y+}} [x^2,0,z^2]\cdot[0,1,0]\,dA = \iint\limits_{S_{y+}} 0\,dA = 0$$

$$\iint\limits_{S_{z-}} \mathbf{F}\cdot\mathbf{n}_{z-}\,dA = \iint\limits_{S_{z-}} [x^2,0,z^2]\cdot[0,0,-1]\,dA = \iint\limits_{S_{z-}} -z^2\,dA \overset{z=0}{=} \iint\limits_{S_{z-}} 0\,dA = 0$$

$$\iint\limits_{S_{z+}} \mathbf{F}\cdot\mathbf{n}_{z+}\,dA = \iint\limits_{S_{z+}} [x^2,0,z^2]\cdot[0,0,1]\,dA = \iint\limits_{S_{z+}} z^2\,dA \overset{z=2}{=} \iint\limits_{S_{z+}} 4\,dA = 4Area(S_{z+}) = 4(12) = 48$$

$$\implies \quad Flux = -12 + 12 + 0 + 0 + 0 + 48 = 48$$

Fortunately, just like Green's Theorem, there is a relationship between the boundary and the enclosed solid that makes this easier.

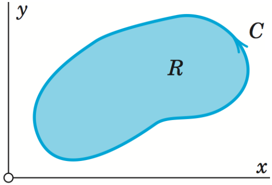
II. Divergence Theorem

Recall: Green's Thm relates a planar closed curve with the region it bounds

The Divergence Theorem relates a closed surface with the solid it bounds

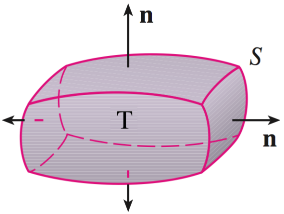
<u>Thm</u>: Let $\mathbf{F} = [M(x,y,z), N(x,y,z), P(x,y,z)]$ be a vector field that has continuous 1st partial derivatives. Also, let T be a bounded volume with boundary S (oriented outward) which is imbedded in $\mathbf{F}$. Then

$$\iint_S \mathbf{F}\cdot\mathbf{n}\,dA = \iiint_T \operatorname{div}\mathbf{F}\,dV = \iiint_T \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z}\right) dV = \iiint_T \left(M_x + N_y + P_z\right) dV$$



Green's Theorem

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \iint_R (\operatorname{curl}\mathbf{F})\cdot\mathbf{k}\,dA$$



Divergence Theorem

$$\iint_S \mathbf{F}\cdot\mathbf{n}\,dA = \iiint_T \operatorname{div}\mathbf{F}\,dV$$

Back to our first problem:

Ex: Compute the flux of $\mathbf{F} = [x^2, 0, z^2]$ through the rectangular box: $|x| \le 1$, $|y| \le 3$, and $0 \le z \le 2$.

$\operatorname{div} \mathbf{F} = (x^2)_x + (0)_y + (z^2)_z = 2x + 2z$

$$Flux = \iint_S \mathbf{F} \cdot \mathbf{n}\, dA = \iiint_T \operatorname{div}\mathbf{F}\, dV = \iiint_T 2x + 2z\, dV$$

$$= \int_0^2 \int_{-3}^3 \int_{-1}^1 2x + 2z\, dx\, dy\, dz$$

$$= 6 \int_0^2 \int_{-1}^1 2x + 2z\, dx\, dz$$

$$= 6 \int_0^2 (x^2 + 2xz)\Big|_{-1}^1\, dz$$

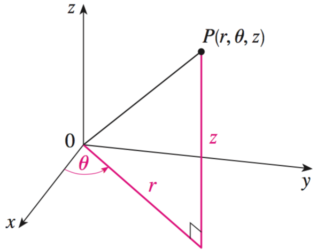
$$= 6 \int_0^2 (1 + 2z) - (1 - 2z)\, dz$$

$$= 6 \int_0^2 4z\, dz = 6\, (2z^2)\Big|_0^2 = 6(8 - 0) = 48$$

III. Review: Cylindrical & Spherical Coordinates

Since we will be computing triple integrals, we should review the different types of coordinates that we may use.

Cylindrical Coordinates:



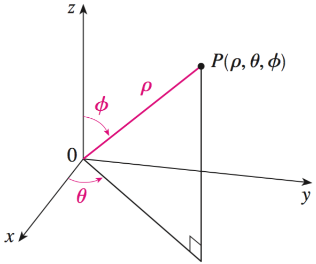
$x = r\cos\theta$

$y = r\sin\theta$

$z = z$

- $x^2 + y^2 = r^2$
- $dV = r\,dr\,d\theta\,dz$

Spherical Coordinates:



$x = \rho\sin\phi\cos\theta$

$y = \rho\sin\phi\sin\theta$

$z = \rho\cos\phi$

- $x^2 + y^2 + z^2 = \rho^2$
- $dV = \rho^2\sin\phi\,d\rho\,d\theta\,d\phi$

Ex: Evaluate $\iint\limits_S \mathbf{F}\cdot\mathbf{n}\,dA$ if $F = [x^3 - y^3, y^3 - z^3, z^3 - x^3]$ and S is given by: $x^2 + y^2 + z^2 \leq 25, \ \ z \geq 0$

Since S is closed, lets apply the Divergence Theorem

$$\text{div}\,\mathbf{F} = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2)$$

$$\iint\limits_S \mathbf{F}\cdot\mathbf{n}\,dA = \iiint\limits_T \text{div}\mathbf{F}\,dV = \iiint\limits_T 3(x^2 + y^2 + z^2)\,dV = (*)$$

Now we need to choose our coordinates. Since S is a sphere, we should use spherical coordinates.

$x = \rho\sin\phi\cos\theta \qquad 0 \leq \rho \leq 5$

$y = \rho\sin\phi\sin\theta \qquad 0 \leq \theta \leq 2\pi$

$z = \rho\cos\phi \qquad 0 \leq \phi \leq \pi$

Remember:

- $x^2 + y^2 + z^2 = \rho^2$
- $dV = \rho^2\sin\phi\,d\rho\,d\theta\,d\phi$

$$(*) = \iiint\limits_T 3(\rho^2)\,dV = \int_0^{\pi}\int_0^{2\pi}\int_0^{5} 3(\rho^2)\,\rho^2\sin\phi\,d\rho\,d\theta\,d\phi = 3\int_0^{\pi}\int_0^{2\pi}\int_0^{5} \rho^4\,\sin\phi\,d\rho\,d\theta\,d\phi$$

$$= 3\int_0^{\pi}\sin\phi\,d\phi\int_0^{2\pi}d\theta\int_0^{5}\rho^4\,d\rho$$

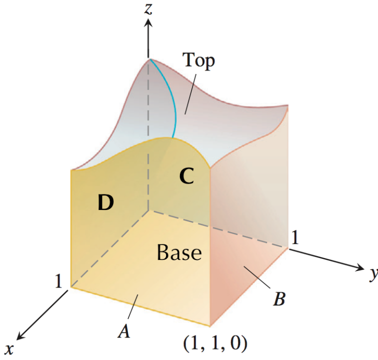
$$= 3\left(-\cos\phi\Big|_{\phi=0}^{\phi=\pi}\right)\left(\theta\Big|_{\theta=0}^{\theta=2\pi}\right)\left(\frac{1}{5}\rho^5\Big|_{\rho=0}^{\rho=5}\right)$$

$$= 3(1+1)(2\pi - 0)\left(\frac{1}{5}(5^5 - 0)\right)$$

$$= 12\pi\cdot 5^4$$

Ex: The the closed cube-like surface, S, shown below has a top that is an unknown smooth surface. Let $\mathbf{F} = x\mathbf{i} - 2y\mathbf{j} + (z+3)\mathbf{k}$ and suppose the outward flux through side A is 1 and through side B is - 3. Find the outward flux through the top?

Let S be the total surface and T the solid bounded by S



Want: $\displaystyle\iint_{Top} \mathbf{F}\cdot\mathbf{n}\,dA = ?$

We are given:

1. $\displaystyle\iint_{A} \mathbf{F}\cdot\mathbf{n}\,dA = 1$
2. $\displaystyle\iint_{B} \mathbf{F}\cdot\mathbf{n}\,dA = -3$

By the Divergence Theorem we also know

$$\iint_{S} \mathbf{F}\cdot\mathbf{n}\,dA = \iiint_{T} \operatorname{div}\mathbf{F}\,dV$$
$$= \iiint_{T} 1-2+1\,dV = \iiint_{T} 0\,dV = 0$$

Since the flux through S is equal to the flux through all of its sides, we have

$$\iint_{S} \mathbf{F}\cdot\mathbf{n}\,dA = \iint_{Top} \mathbf{F}\cdot\mathbf{n}\,dA + \iint_{Base} \mathbf{F}\cdot\mathbf{n}\,dA + \iint_{A} \mathbf{F}\cdot\mathbf{n}\,dA + \iint_{B} \mathbf{F}\cdot\mathbf{n}\,dA + \iint_{C} \mathbf{F}\cdot\mathbf{n}\,dA + \iint_{D} \mathbf{F}\cdot\mathbf{n}\,dA$$

$$0 = \iint_{Top} \mathbf{F}\cdot\mathbf{n}\,dA + \iint_{Base} \mathbf{F}\cdot\mathbf{n}\,dA + 1 + -3 + \iint_{C} \mathbf{F}\cdot\mathbf{n}\,dA + \iint_{D} \mathbf{F}\cdot\mathbf{n}\,dA \quad (*)$$

For the Base, we know: $\mathbf{n} = [0, 0, -1]$ and $z = 0$

$$\implies \mathbf{F} \cdot \mathbf{n} = [x, -2y, z-3] \cdot [0, 0, -1] = z - 3 = -3$$

$$\implies \iint_{Base} \mathbf{F} \cdot \mathbf{n}\, dA = \iint_{Base} -3\, dA = -3 \cdot \text{Area}(Base) = -3$$

For Side C, we know: $\mathbf{n} = [-1, 0, 0]$ and $x = 0$

$$\implies \mathbf{F} \cdot \mathbf{n} = [x, -2y, z-3] \cdot [-1, 0, 0] = -x = 0$$

$$\implies \iint_{C} \mathbf{F} \cdot \mathbf{n}\, dA = \iint_{C} 0\, dA = 0$$

For Side D, we know: $\mathbf{n} = [0, -1, 0]$ and $y = 0$

$$\implies \mathbf{F} \cdot \mathbf{n} = [x, -2y, z-3] \cdot [0, -1, 0] = 2y = 0$$

$$\implies \iint_{D} \mathbf{F} \cdot \mathbf{n}\, dA = \iint_{D} 0\, dA = 0$$

Updating (*), we have:

$$0 = \iint_{Top} \mathbf{F} \cdot \mathbf{n}\, dA + \quad -3 \quad + \quad 1 \quad + \quad -3 \quad + \quad 0 \quad + \quad 0$$

Therefore, $\displaystyle\iint_{Top} \mathbf{F} \cdot \mathbf{n}\, dA = 5$